Exp No:14 Experiment to understand feature scaling.

AIM:

To understand the importance of feature scaling.

CODE:

```
from google.colab import files
files.upload()
import numpy as np
import pandas as pd

df = pd.read_csv('/content/pre-process_datasample.csv')
df

df.Country.fillna(df.Country.mode()[0], inplace=True)

features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values

from sklearn.impute import SimpleImputer

age = SimpleImputer(strategy="mean", missing_values=np.nan)
Salary = SimpleImputer(strategy="mean", missing_values=np.nan)

age.fit(features[:, [1]])
Salary.fit(features[:, [2]])

features[:, [1]] = age.transform(features[:, [1]])
features[:, [2]] = Salary.transform(features[:, [2]])

features

from sklearn.preprocessing import OneHotEncoder

oh = OneHotEncoder(sparse_output=False)
Country = oh.fit_transform(features[:, [0]])

Country

final_set = np.concatenate((Country, features[:, [1, 2]]), axis=1)
final_set

from sklearn.preprocessing import StandardScaler
```

```
sc = StandardScaler()
sc.fit(final_set)

feat_standard_scaler = sc.transform(final_set)
feat_standard_scaler

from sklearn.preprocessing import MinMaxScaler

mms = MinMaxScaler(feature_range=(0, 1))
mms.fit(final_set)

feat_minmax_scaler = mms.transform(final_set)
feat_minmax_scaler
```

OUTPUT:

```
  df.Country.fillna(df.Country.mode()[0], inplace=True)
array([[1.        , 0.        , 0.        , 0.73913043, 0.68571429],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 1.        , 0.        , 0.13043478, 0.17142857],
       [0.        , 0.        , 1.        , 0.47826087, 0.37142857],
       [0.        , 1.        , 0.        , 0.56521739, 0.45079365],
       [1.        , 0.        , 0.        , 0.34782609, 0.28571429],
       [0.        , 0.        , 1.        , 0.51207729, 0.11428571],
       [1.        , 0.        , 0.        , 0.91304348, 0.88571429],
       [1.        , 0.        , 0.        , 1.        , 1.        ],
       [1.        , 0.        , 0.        , 0.43478261, 0.54285714]])
```

RESULT:

Thus,the given data was analyzed successfully and required output was obtained.